Grade 10 - Term 3
Midterm Solution

Contents

Evaluated criteria

- C1. Online Coffee Sales Investment.
- C2. Three-Scenario Inventory Investment.
- C3. Small-Business Investment.
- C4. Drawing and comparing probability distributions.
- C5. Uses the standard normal distribution and its tables to calculate probabilities.
- C6. Uses normal distributions in contextual situations.
- C7. Uses inverse normal reasoning to find values and intervals.

1 (9) Online Coffee Sales Investment (C1)

A small entrepreneur runs a specialty coffee resale business. Let V be the end-of-quarter value, measured in million pesos. The final value cannot be below 20 and cannot exceed 68. Lower values are possible, but higher values become gradually more likely across this interval, so the final value is modeled with an increasing density. The initial investment is 32 million pesos. The probability of the extreme value of 20 million pesos can be considered to be 0.

Questions/Tasks.

- **a)** **(3)** Determine the density constant and write the pdf.
- **b)** **(2)** Find the probability of loss and the probability of earning.
- **c)** **(4)** Compute the probability that the money obtained falls between 32 and 56 million.

a) (3) Determine the density constant and write the pdf.

Let V denote the end-of-quarter value, measured in million pesos. Since the density rises linearly from the lower endpoint 20 to the upper endpoint 68, we model it with a linear function.

One possible general form for a line is

$$f(v) = m(v + c).$$

We know that the density starts at 0 when $v = 20$, so

$$f(20) = m(20 + c) = 0.$$

Since the density is increasing, $m \neq 0$. Therefore,

$$20 + c = 0 \qquad \longrightarrow \qquad c = -20 \qquad \longrightarrow \qquad f(v) = m(v - 20)$$

Since the region under the density must have total area 1, and the graph forms a triangle with base and height:

$$B = 68 - 20 = 48 \qquad\qquad h = f(68)$$

we have

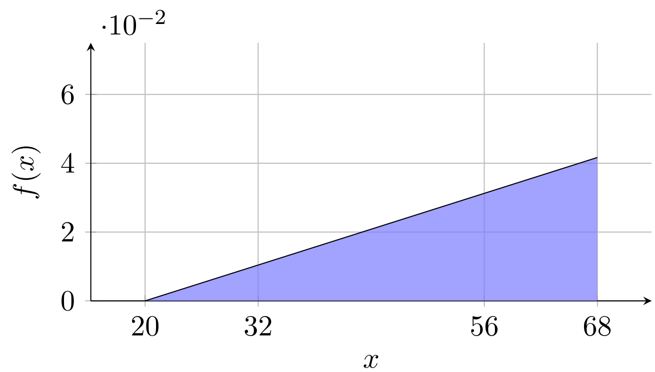
$$A_\triangle = \frac{1}{2}Bh = \frac{1}{2}(48)h = 1 \qquad \longrightarrow \qquad 24h = 1 \qquad \longrightarrow \qquad h = \frac{1}{24}.$$

Since $f(68) = h$, we now use $h = \frac{1}{24}$. Using $f(v) = m(v - 20)$, substitute $v = 68$:

$$f(68) = m(68 - 20) \qquad \longrightarrow \qquad \frac{1}{24} = h = f(68) = 48m \qquad \longrightarrow \qquad \frac{1}{24} = 48m \qquad \longrightarrow \qquad m = \frac{1}{24 \cdot 48} = \frac{1}{1152}$$

Hence the pdf is

$$f(v) = \begin{cases} \dfrac{1}{1152}(v - 20), & 20 \le v \le 68, \\ 0, & \text{otherwise.} \end{cases}$$



b) (2) Find the probability of loss and the probability of earning.

Probability of loss. A loss occurs when the final value is below the purchase price of 32 million pesos. Therefore, we need to find

$$P(\text{loss}) = P(V < 32)$$

On the interval from 20 to 32, the graph forms a smaller triangle. Its base and height are

$$B = 32 - 20 = 12 \qquad\qquad h = f(32)$$

Using the pdf from part a, we calculate the height:

$$h = f(32) = \frac{1}{1152}(32 - 20) = \frac{12}{1152} = \frac{1}{96}$$

Since the probability is the area under the density curve, we have

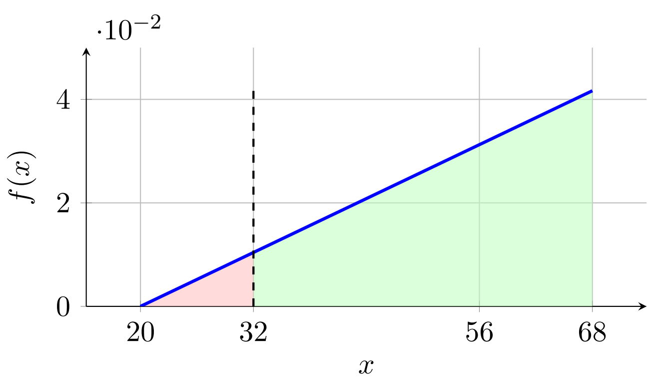
$$P(\text{loss}) = P(V < 32) = A_{\triangle} = \frac{1}{2}Bh = \frac{1}{2}(12)\left(\frac{1}{96}\right) = \frac{6}{96} = \frac{1}{16}$$

Probability of earning. An earning occurs when the final value is above the purchase price of 32 million pesos. Therefore, we need to find $P(\text{earning})$, we proceed using complement rule

$$P(\text{earning}) = P(V > 32) = 1 - P(V < 32) = 1 - \frac{1}{16} = \frac{15}{16}$$

Therefore,

$$\text{Loss} \to P(V < 32) = \frac{1}{16} \qquad\qquad \text{Earning} \to P(V > 32) = \frac{15}{16}$$



c) (4) Compute the probability that the money obtained falls between 32 and 56 million.

Now compute the probability that the money obtained falls between 32 and 56 million pesos. Therefore, we need to find

$$P(32 \leq V \leq 56)$$

Solution 1: Trapezoid method.

On the interval from 32 to 56, the region under the density forms a trapezoid. Its base and heights are

$$B = 56 - 32 = 24 \qquad\qquad h_1 = f(32) \qquad\qquad h_2 = f(56)$$

Using the pdf from part a, we calculate the first height:

$$h_1 = f(32) = \frac{1}{1152}(32 - 20) = \frac{12}{1152} = \frac{1}{96}$$

Using the same pdf, we calculate the second height:

$$h_2 = f(56) = \frac{1}{1152}(56 - 20) = \frac{36}{1152} = \frac{1}{32}$$

Since the probability is the area under the density curve, we have

$$\begin{aligned} P(32 \le V \le 56) = A_{\text{trapezoid}} &= \frac{1}{2}B(h_1 + h_2) \\ &= \frac{1}{2}(24)\left(\frac{1}{96} + \frac{1}{32}\right) = 12\left(\frac{1}{96} + \frac{3}{96}\right) \\ &= 12\left(\frac{4}{96}\right) = \frac{48}{96} = \frac{1}{2} \end{aligned}$$

Solution 2: Larger triangle minus smaller triangle.

First, compute the probability below 56. On the interval from 20 to 56, the graph forms a triangle. Its base and height are

$$B_1 = 56 - 20 = 36 \qquad\qquad h_1 = f(56)$$

Using the pdf from part a), we calculate the height:

$$h_1 = f(56) = \frac{1}{1152}(56 - 20) = \frac{36}{1152} = \frac{1}{32}$$

Therefore,

$$P(V < 56) = A_{\triangle} = \frac{1}{2}B_1 h_1 = \frac{1}{2}(36)\left(\frac{1}{32}\right) = \frac{18}{32} = \frac{9}{16}$$

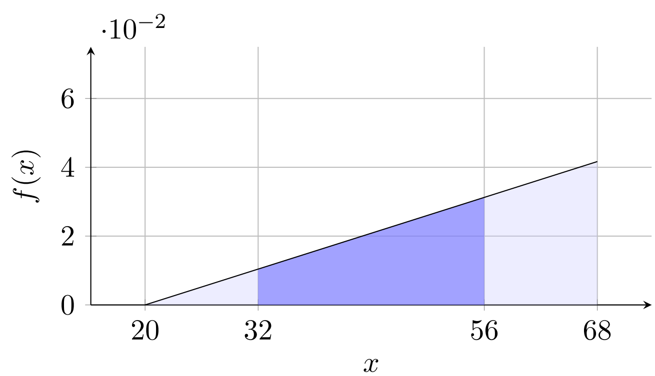
The probability below 32 was already calculated in part b), so we use

$$P(V < 32) = \frac{1}{16}$$

Since the region from 32 to 56 is obtained by subtracting the smaller triangle from the larger triangle, we have

$$P(32 \le V \le 56) = P(V < 56) - P(V < 32) = \frac{9}{16} - \frac{1}{16} = \frac{8}{16} = \frac{1}{2}$$

Thus, the probability that the money obtained falls between 32 and 56 million pesos is 50%



2 (9) Three-Scenario Inventory Investment (C2)

Problem description. A seller invests in a short-term inventory deal to resell packaged foods in nearby cities. For the quarter, they consider 3 realistic market scenarios. In Scenario 1, transport bottlenecks continue and restocking stays slow; this has probability 0.2. In Scenario 2, transport returns to normal and inventory rotates faster; this has probability 0.5. In Scenario 3, demand increases because restaurants and small stores begin buying in bulk; this has probability 0.3. If Scenario 1 occurs, the final value of the investment is expected to end between 48 and 68 million pesos, with all values in that interval treated as equally plausible. If Scenario 2 occurs, the final value is expected to end between 54 and 104 million pesos, again with all values in that interval treated as equally plausible. If Scenario 3 occurs, the final value is expected to end between 60 and 90 million pesos, again with all values in that interval treated as equally plausible. The initial investment is 60 million pesos.

Questions/Tasks.

a) **(2)** Write the conditional density for each scenario.

b) **(3)** Build the unconditional pdf of the final value as a piecewise function.

c) **(4)** Find the probability of loss and the probability of earning.

A conditional density function describes how the final value is distributed once one specific scenario is assumed to be true. In this problem, each scenario has its own density because the investment behaves differently depending on market conditions.

The unconditional density function combines the scenario-specific densities into one overall model. It is obtained by weighting each conditional density by the probability of its scenario. The overlap region is important because it contains values that are plausible under at least two market scenarios.

a) (2) Write the conditional density for each scenario.

Let V be the final value of the investment, measured in million pesos.

Under Scenario 1, the value is uniformly distributed on $[48, 68]$. Therefore, the conditional density is

$$f_{V|S_1}(v) = \begin{cases} \dfrac{1}{20}, & 48 \le v \le 68, \\ 0, & \text{otherwise.} \end{cases}$$

Under Scenario 2, the value is uniformly distributed on $[54, 104]$. Therefore, the conditional density is

$$f_{V|S_2}(v) = \begin{cases} \dfrac{1}{50}, & 54 \le v \le 104, \\ 0, & \text{otherwise.} \end{cases}$$

Under Scenario 3, the value is uniformly distributed on $[60, 90]$. Therefore, the conditional density is

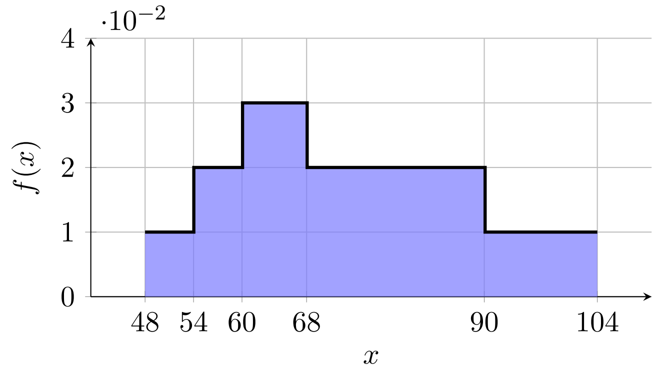
$$f_{V|S_3}(v) = \begin{cases} \dfrac{1}{30}, & 60 \le v \le 90, \\ 0, & \text{otherwise.} \end{cases}$$

The scenario probabilities are

$$P(S_1) = 0.2 \qquad P(S_2) = 0.5 \qquad P(S_3) = 0.3.$$

So the unconditional density is the mixture

$$f_V(v) = 0.2\, f_{V|S_1}(v) + 0.5\, f_{V|S_2}(v) + 0.3\, f_{V|S_3}(v).$$



b) (3) Build the unconditional pdf of the final value as a piecewise function.

Now build it by intervals.

For $48 \le v < 54$, only Scenario 1 contributes, so

$$f_V(v) = 0.2\left(\frac{1}{20}\right) = \frac{1}{100}.$$

For $54 \le v < 60$, Scenarios 1 and 2 contribute, so

$$f_V(v) = 0.2\left(\frac{1}{20}\right) + 0.5\left(\frac{1}{50}\right) = \frac{1}{100} + \frac{1}{100} = \frac{1}{50}.$$

For $60 \le v \le 68$, all three scenarios contribute, so

$$f_V(v) = 0.2\left(\frac{1}{20}\right) + 0.5\left(\frac{1}{50}\right) + 0.3\left(\frac{1}{30}\right) = \frac{1}{100} + \frac{1}{100} + \frac{1}{100} = \frac{3}{100}.$$

For $68 < v \le 90$, Scenarios 2 and 3 contribute, so

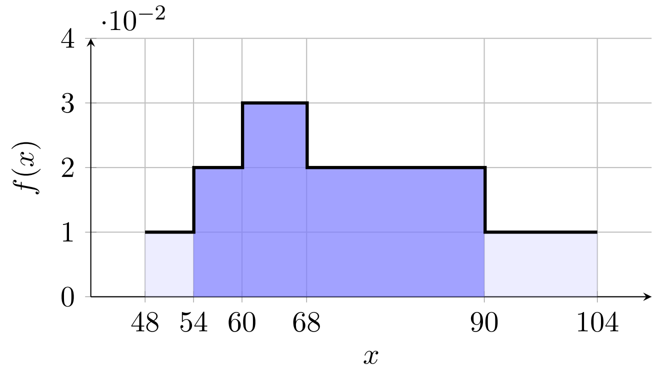
$$f_V(v) = 0.5\left(\frac{1}{50}\right) + 0.3\left(\frac{1}{30}\right) = \frac{1}{100} + \frac{1}{100} = \frac{1}{50}.$$

For $90 < v \le 104$, only Scenario 2 contributes, so

$$f_V(v) = 0.5\left(\frac{1}{50}\right) = \frac{1}{100}.$$

Therefore, the unconditional pdf is

$$f_V(v) = \begin{cases} \dfrac{1}{100}, & 48 \le v < 54, \\ \dfrac{1}{50}, & 54 \le v < 60, \\ \dfrac{3}{100}, & 60 \le v \le 68, \\ \dfrac{1}{50}, & 68 < v \le 90, \\ \dfrac{1}{100}, & 90 < v \le 104, \\ 0, & \text{otherwise.} \end{cases}$$



c) (4) Find the probability of loss and the probability of earning.

Probability of loss. A loss occurs when the final value is below the initial investment of 60 million pesos. This includes the interval from 48 to 54, plus the interval from 54 to 60.

From 48 to 54, the area is

$$6\left(\frac{1}{100}\right) = \frac{3}{50}.$$

From 54 to 60, the width is 6 and the height is $\frac{1}{50}$, so the area is

$$6\left(\frac{1}{50}\right) = \frac{3}{25}.$$

Therefore,

$$P(\text{loss}) = P(V < 60) = \frac{3}{50} + \frac{3}{25} = \frac{9}{50}.$$

Probability of earning. An earning occurs when the final value is above the initial investment of 60 million pesos. Since the total probability is 1 and there is no probability mass at exactly 60, use the complement rule

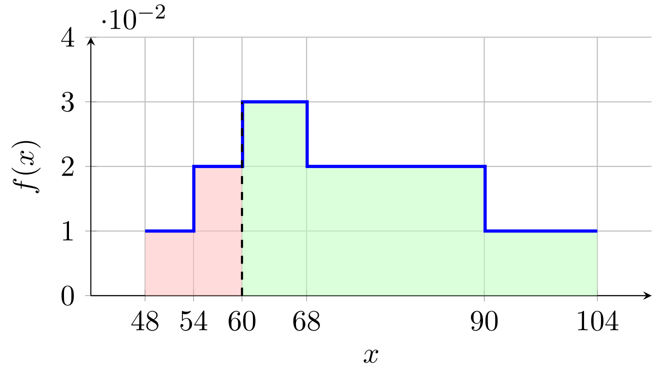
$$P(\text{earning}) = P(V > 60) = 1 - P(V < 60).$$

From the previous result,

$$P(V < 60) = \frac{9}{50}.$$

Therefore,

$$P(V > 60) = 1 - \frac{9}{50} \qquad \longrightarrow \qquad P(V > 60) = \frac{41}{50}.$$



3 (9) Small-Business Investment (C3)

An investor invests in a short-term small-business opportunity: buying low-cost clothing bundles to resell online and at weekend markets. They estimate the final value cannot be below 0 million pesos because, in this simplified model, the end-of-period value cannot be negative. They also estimate it cannot exceed 40 million pesos because their first-round sales capacity is limited. Their most likely final value is 16 million pesos, based on expected demand from regular customers. Values below 16 million pesos are possible, but become less likely as results move farther down from that level. Values above 16 million pesos are also possible, but after that point, high outcomes become gradually less likely. Since the model is continuous, the probability of obtaining any exact extreme value can be considered 0, so the probabilities of ending at exactly 0 million pesos or exactly 40 million pesos are treated as 0. The initial investment is 10 million pesos. A loss occurs if the final value is below that amount, and an earning occurs if the final value is above it.

Questions/Tasks.

- **a)** **(3)** Write the pdf as a piecewise function.
- **b)** **(3)** Find the probability of loss and the probability of earning.
- **c)** **(3)** Find the specific probability of earning more than 50% of the investment.

a) (3) Write the pdf as a piecewise function.

Let V be the final value of the investment, measured in million pesos.

The support is from 0 to 40, and the most likely value is 16. Since the model is triangular, the graph rises linearly from $(0, 0)$ to the peak at $(16, h)$, then falls linearly from $(16, h)$ to $(40, 0)$.

Since the region under the density must have total area 1, and the graph forms a triangle with base and height:

$$B = 40 - 0 = 40 \qquad\qquad h = f(16)$$

we have

$$A_\triangle = \frac{1}{2}Bh = \frac{1}{2}(40)h = 1 \qquad \longrightarrow \qquad 20h = 1 \qquad \longrightarrow \qquad h = \frac{1}{20}.$$

Left-hand piece.

On the interval from 0 to 16, the graph is linear. One possible general form for a line is

$$f(v) = m(v + c)$$

We know that the density starts at 0 when $v = 0$, so

$$f(0) = m(0 + c) = 0 \qquad \longrightarrow \qquad c = 0 \qquad \longrightarrow \qquad f(v) = mv$$

Since $f(16) = h$, we now use $h = \frac{1}{20}$. Substituting $v = 16$:

$$f(16) = 16m \qquad \longrightarrow \qquad \frac{1}{20} = h = f(16) = 16m \qquad \longrightarrow \qquad m = \frac{1}{20 \cdot 16} = \frac{1}{320}.$$

Thus, the left-hand piece is

$$f(v) = \frac{v}{320} \qquad \text{for } 0 \le v \le 16.$$

Right-hand piece.

On the interval from 16 to 40, the graph is also linear. We use again

$$f(v) = m(v + c).$$

We know that the density ends at 0 when $v = 40$, so

$$f(40) = m(40 + c) = 0 \qquad \longrightarrow \qquad 40 + c = 0 \qquad \longrightarrow \qquad c = -40 \qquad \longrightarrow \qquad f(v) = m(v - 40)$$

Since $f(16) = h$, we now use $h = \frac{1}{20}$. Substituting $v = 16$:

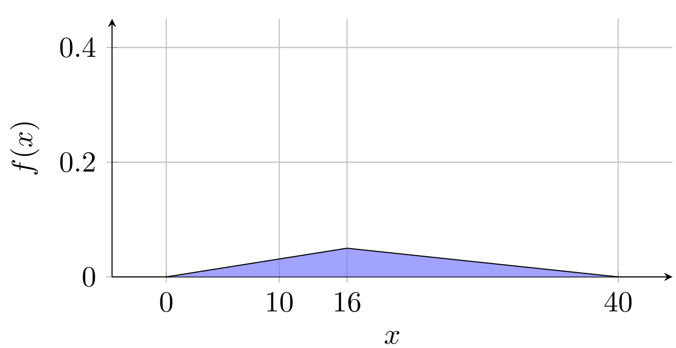
$$f(16) = m(16 - 40) \qquad \longrightarrow \qquad \frac{1}{20} = h = f(16) = -24m \qquad \longrightarrow \qquad m = -\frac{1}{20 \cdot 24} = -\frac{1}{480}.$$

Thus, the right-hand piece is

$$f(v) = -\frac{1}{480}(v - 40) = \frac{40 - v}{480} \qquad \text{for } 16 \le v \le 40.$$

Therefore, the pdf is

$$f(v) = \begin{cases} \dfrac{v}{320}, & 0 \le v \le 16, \\ \dfrac{40 - v}{480}, & 16 \le v \le 40, \\ 0, & \text{otherwise.} \end{cases}$$



b) (3) Find the probability of loss and the probability of earning.

Probability of loss. A loss occurs when the final value is below the purchase price of 10 million pesos. Therefore, we need to find

$$P(\text{loss}) = P(V < 10).$$

On the interval from 0 to 10, the graph forms a triangle. Its base and height are

$$B = 10 - 0 = 10 \qquad\qquad h = f(10)$$

Using the pdf from part a, we calculate the height:

$$h = f(10) = \frac{10}{320} = \frac{1}{32}.$$

Since the probability is the area under the density curve, we have

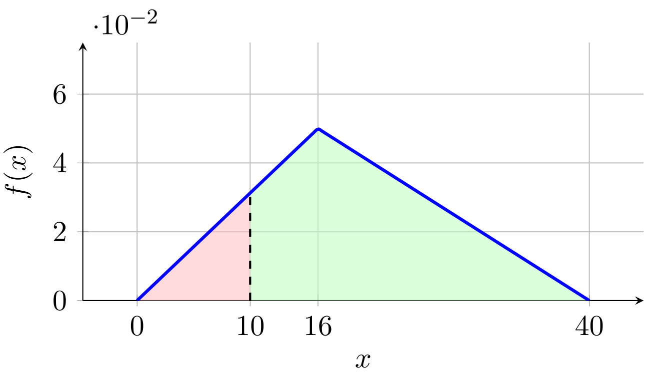
$$P(\text{loss}) = P(V < 10) = A_{\triangle} = \frac{1}{2}Bh = \frac{1}{2}(10)\left(\frac{1}{32}\right) = \frac{5}{32}.$$

Probability of earning. An earning occurs when the final value is above the purchase price of 10 million pesos. Therefore, we need to find $P(\text{earning})$, we proceed using complement rule:

$$P(\text{earning}) = P(V > 10) = 1 - P(V < 10) = 1 - \frac{5}{32} = \frac{27}{32}$$

Therefore,

$$\text{Loss} \to P(V < 10) = \frac{5}{32} \qquad\qquad \text{Earning} \to P(V > 10) = \frac{27}{32}.$$



c) (3) Find the specific probability of earning more than 50% of the investment.

The investment amount is 10 million pesos, so earning more than 50% means ending above

$$10(1.5) = 15.$$

Therefore, we need to find

$$P(\text{earning more than } 50\%) = P(V > 15).$$

Since 15 lies to the left of the most likely value 16, the complement rule is convenient. We first calculate the probability below 15:

$$P(V > 15) = 1 - P(V \le 15).$$

On the interval from 0 to 15, the graph forms a triangle. Its base and height are

$$B = 15 - 0 = 15 \qquad\qquad h = f(15)$$

Using the pdf from part a, we calculate the height:

$$h = f(15) = \frac{15}{320} = \frac{3}{64}.$$

Since the probability is the area under the density curve, we have

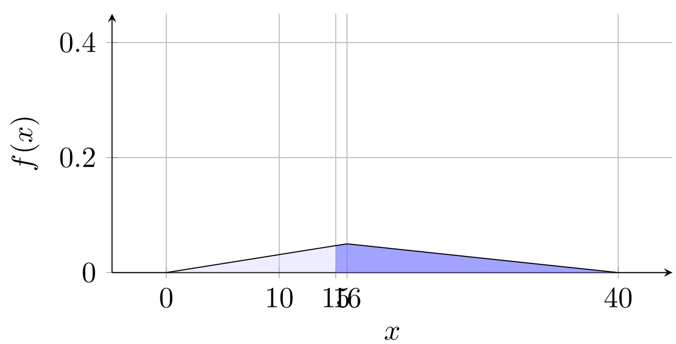
$$P(V \le 15) = A_{\triangle} = \frac{1}{2}Bh = \frac{1}{2}(15)\left(\frac{3}{64}\right) = \frac{45}{128}.$$

Now use the complement rule:

$$P(V > 15) = 1 - P(V \le 15) = 1 - \frac{45}{128} = \frac{83}{128}.$$

Thus, the probability of earning more than 50% of the investment is

$$P(\text{earning more than } 50\%) = P(V > 15) = \frac{83}{128}.$$



4 (9) Probability Distribution Graphs (C4)

Draw the distributions in each part on the same set of axes. Label graphs A–D clearly.

Questions/Tasks.

a) **(3)** Draw four uniform distributions on the same graph.

b) **(3)** Draw four centered triangular distributions on the same graph.

c) **(3)** Draw four normal distributions on the same graph.

a) (3) Draw four uniform distributions on the same graph.

Draw the four uniform distributions on one set of axes.

- Graph A: $\mu = -1$, $\sigma \approx 0.58$
- Graph B: $\mu = 2$, $\sigma \approx 1.15$
- Graph C: $\mu = 6$, $\sigma \approx 1.73$
- Graph D: $\mu = 9$, $\sigma \approx 2.31$

Solution. For a uniform distribution $U[a, b]$:

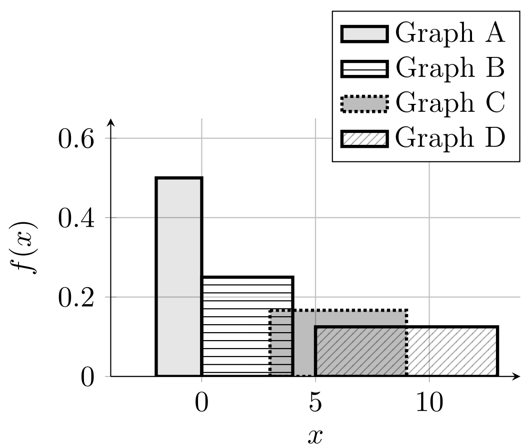
$$\mu = \frac{a+b}{2}, \qquad \sigma = \frac{b-a}{\sqrt{12}}.$$

So

$$a = \mu - \frac{\sigma\sqrt{12}}{2}, \qquad b = \mu + \frac{\sigma\sqrt{12}}{2}.$$

Using each pair (μ, σ):

$$A : U[-2, 0], \qquad B : U[0, 4], \qquad C : U[3, 9], \qquad D : U[5, 13].$$



Final answer:

$$A : U[-2, 0], \quad B : U[0, 4], \quad C : U[3, 9], \quad D : U[5, 13].$$

b) (3) Draw four centered triangular distributions on the same graph.

Draw the four centered triangular distributions on one set of axes.

- Graph A: $\mu = -1$, $\sigma \approx 0.82$
- Graph B: $\mu = 2$, $\sigma \approx 1.22$
- Graph C: $\mu = 6$, $\sigma \approx 0.41$
- Graph D: $\mu = 9$, $\sigma \approx 1.63$

Solution. For a centered triangular distribution (a, c, b), we have

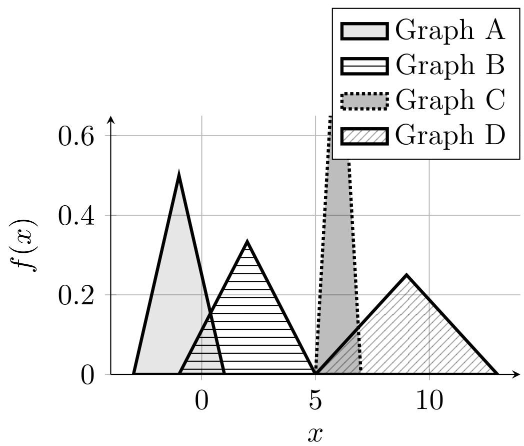
$$c = \mu, \qquad a = \mu - d, \qquad b = \mu + d, \qquad \sigma = \frac{d}{\sqrt{6}}.$$

Then

$$d = \sigma\sqrt{6}, \qquad a = \mu - \sigma\sqrt{6}, \qquad b = \mu + \sigma\sqrt{6}.$$

Using each pair (μ, σ):

$$A : (-3, -1, 1), \qquad B : (-1, 2, 5), \qquad C : (5, 6, 7), \qquad D : (5, 9, 13).$$



Final answer:

$$A : (-3, -1, 1), \quad B : (-1, 2, 5), \quad C : (5, 6, 7), \quad D : (5, 9, 13).$$

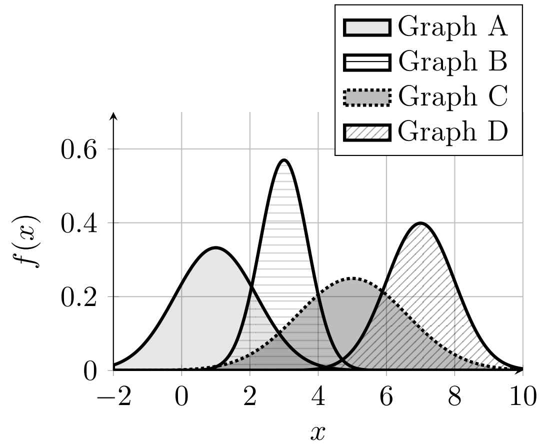
c) (3) Draw four normal distributions on the same graph.

Draw the following four normal distributions on the same set of axes. Clearly distinguish the graphs in the picture.

- Graph A: $X \sim N(1.0, 1.2^2)$
- Graph B: $Y \sim N(3.0, 0.7^2)$
- Graph C: $Z \sim N(5.0, 1.6^2)$
- Graph D: $W \sim N(7.0, 1.0^2)$

Solution. Each normal graph is determined by its mean (center) and standard deviation (spread).

- A: centered at 1.0 with spread $\sigma = 1.2$.
- B: centered at 3.0 with smallest spread $\sigma = 0.7$ (tallest curve).
- C: centered at 5.0 with largest spread $\sigma = 1.6$ (flattest curve).
- D: centered at 7.0 with spread $\sigma = 1.0$.



Final answer:

$$N(1.0, 1.2^2), \quad N(3.0, 0.7^2), \quad N(5.0, 1.6^2), \quad N(7.0, 1.0^2).$$

Standard normal table (values of $\Phi(z)$).

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

5 (10) Standard Normal Table Problems (C5)

Let $Z \sim N(0, 1)$. Let $\Phi(z) = P(Z < z)$.

This section prepares the standard-normal values used later in C6 and C7.

Questions/Tasks.

a) **(4) Direct and symmetry-based table probabilities:** $P(Z < 1.50)$, $P(Z < -1.20)$, $P(Z < 0.25)$, $P(Z < 1.15)$, $P(Z < -1.05)$, and $P(Z < 1.35)$.

b) **(2) Complement-rule probabilities:** $P(Z > 1.50)$, $P(Z > -0.80)$, and $P(Z > 1.35)$.

c) **(4) Interval and tail-union probabilities:** $P(0.25 < Z < 1.15)$, $P(Z < -1.05 \text{ or } Z > 1.35)$, $P(Z < -1.28)$, $P(Z < 0.84)$, $P(Z > 1.28)$, $P(Z > -0.84)$, $P(-1.96 < Z < 1.96)$, and $P(Z < -1.64 \text{ or } Z > 1.64)$.

a) (4) Direct and symmetry-based table probabilities

Find the following left-tail probabilities. These values are used later when contextual values are converted to Z-scores in C6.

$$P(Z < 1.50) \qquad P(Z < -1.20) \qquad P(Z < 0.25) \qquad P(Z < 1.15)$$
$$P(Z < -1.05) \qquad P(Z < 1.35)$$

For positive values, read $\Phi(z)$ directly from the table:

$$P(Z < 1.50) = \Phi(1.50) = 0.9332$$

$$P(Z < 0.25) = \Phi(0.25) = 0.5987$$

$$P(Z < 1.15) = \Phi(1.15) = 0.8749$$

$$P(Z < 1.35) = \Phi(1.35) = 0.9115$$

For negative values, use symmetry:

$$\Phi(-x) = 1 - \Phi(x)$$

$$P(Z < -1.20) = \Phi(-1.20) = 1 - \Phi(1.20) = 1 - 0.8849 = 0.1151$$

$$P(Z < -1.05) = \Phi(-1.05) = 1 - \Phi(1.05) = 1 - 0.8531 = 0.1469$$

b) (2) Complement-rule probabilities

Find the following right-tail probabilities. These values support the right-tail calculations in C6.

$$P(Z > 1.50) \qquad P(Z > -0.80) \qquad P(Z > 1.35)$$

For positive boundaries:

$$P(Z > z) = 1 - \Phi(z)$$

$$P(Z > 1.50) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668$$

$$P(Z > 1.35) = 1 - \Phi(1.35) = 1 - 0.9115 = 0.0885$$

For the negative boundary:

$$P(Z > -0.80) = 1 - \Phi(-0.80)$$

$$\Phi(-0.80) = 1 - \Phi(0.80)$$

$$P(Z > -0.80) = 1 - (1 - \Phi(0.80)) = \Phi(0.80)$$

$$P(Z > -0.80) = 0.7881$$

c) (4) Interval and tail-union probabilities

Compute each requested probability from the exam list.

$$P(0.25 < Z < 1.15) = \Phi(1.15) - \Phi(0.25) = 0.8749 - 0.5987 = 0.2762$$

$$P(Z < -1.05 \text{ or } Z > 1.35) = P(Z < -1.05) + P(Z > 1.35)$$

$$P(Z < -1.05) = 1 - \Phi(1.05) = 1 - 0.8531 = 0.1469, \quad P(Z > 1.35) = 1 - \Phi(1.35) = 0.0885$$

$$P(Z < -1.05 \text{ or } Z > 1.35) = 0.1469 + 0.0885 = 0.2354$$

$$P(Z < -1.28) = 1 - \Phi(1.28) = 1 - 0.8997 = 0.1003, \qquad P(Z < 0.84) = \Phi(0.84) = 0.7995$$

$$P(Z > 1.28) = 1 - \Phi(1.28) = 0.1003, \qquad P(Z > -0.84) = 1 - \Phi(-0.84) = \Phi(0.84) = 0.7995$$

$$P(-1.96 < Z < 1.96) = \Phi(1.96) - \Phi(-1.96) = 0.9750 - 0.0250 = 0.9500$$

$$P(Z < -1.64 \text{ or } Z > 1.64) = P(Z < -1.64) + P(Z > 1.64)$$

$$P(Z < -1.64) = 1 - \Phi(1.64) = 1 - 0.9495 = 0.0505, \quad P(Z > 1.64) = 1 - \Phi(1.64) = 0.0505$$

$$P(Z < -1.64 \text{ or } Z > 1.64) = 0.0505 + 0.0505 = 0.1010$$

6 (10) Contextual Normal Distribution Problems (C6)

Use

$$Z = \frac{X - \mu}{\sigma}$$

to convert each contextual value into a standard normal value.

Questions/Tasks.

a) **(2) Exam scores.** If exam scores follow $X \sim N(70, 8^2)$, find the probability that a student scores below 82 and the probability that a student scores below 60.4: $P(X < 82)$ and $P(X < 60.4)$.

b) **(2) Delivery times.** If delivery times follow $X \sim N(56, 5^2)$, find the probability that a delivery takes more than 63.5 minutes and the probability that a delivery takes more than 52 minutes: $P(X > 63.5)$ and $P(X > 52)$.

c) **(3) Weekly app usage.** If weekly app usage follows $X \sim N(14, 4^2)$, find the probability that weekly app usage is between 15 and 18.6 hours: $P(15 < X < 18.6)$.

d) **(3) Scholarship scores.** If scholarship scores follow $X \sim N(300, 20^2)$, find the probability that a score is below 279 or above 327: $P(X < 279 \text{ or } X > 327)$.

a) (2) Exam scores

Exam scores follow

$$X \sim N(70, 8^2)$$

First probability. Find $P(X < 82)$.

$$P(X < 82) = P\left(Z < \frac{82 - 70}{8}\right) = P(Z < 1.50)$$

$$P(X < 82) = \Phi(1.50) = 0.9332$$

Second probability. Find $P(X < 60.4)$.

$$P(X < 60.4) = P\left(Z < \frac{60.4 - 70}{8}\right) = P(Z < -1.20)$$

$$P(X < 60.4) = \Phi(-1.20) = 0.1151$$

This uses the same table logic as C5 part a).

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b) (2) Delivery times

Delivery times follow

$$X \sim N(56, 5^2)$$

First probability. Find $P(X > 63.5)$.

$$P(X > 63.5) = P\left(Z > \frac{63.5 - 56}{5}\right) = P(Z > 1.50)$$

$$P(X > 63.5) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668$$

Second probability. Find $P(X > 52)$.

$$P(X > 52) = P\left(Z > \frac{52 - 56}{5}\right) = P(Z > -0.80)$$

$$P(X > 52) = 1 - \Phi(-0.80) = 1 - 0.2119 = 0.7881$$

This uses the complement-rule logic from C5 part b).

c) (3) Weekly app usage

Weekly app usage follows

$$X \sim N(14, 4^2)$$

Find $P(15 < X < 18.6)$.

$$P(15 < X < 18.6) = P\left(\frac{15 - 14}{4} < Z < \frac{18.6 - 14}{4}\right)$$
$$P(15 < X < 18.6) = P(0.25 < Z < 1.15)$$
$$P(15 < X < 18.6) = \Phi(1.15) - \Phi(0.25) = 0.8749 - 0.5987 = 0.2762$$

This uses the interval/tail logic from C5 part c).

d) (3) Scholarship scores

Scholarship scores follow

$$X \sim N(300, 20^2)$$

Find $P(X < 279 \text{ or } X > 327)$.

$$P(X < 279 \text{ or } X > 327) = P\left(Z < \frac{279 - 300}{20}\right) + P\left(Z > \frac{327 - 300}{20}\right)$$
$$P(X < 279 \text{ or } X > 327) = P(Z < -1.05) + P(Z > 1.35)$$
$$P(X < 279 \text{ or } X > 327) = \Phi(-1.05) + 1 - \Phi(1.35) = 0.1469 + 1 - 0.9115 = 0.2354$$

This uses the interval/tail logic from C5 part c).

7 (10) Inverse Normal Problems (C7)

Use

$$Z = \frac{X - \mu}{\sigma} \qquad \text{and} \qquad x = \mu + \sigma z$$

to convert between standard normal values and contextual values.

Questions/Tasks.

a) **(2) Lower-than cutoffs.** If commute times follow $X \sim N(38, 7^2)$, find the cutoff below which the lowest 10% of commutes fall and the cutoff below which the lowest 80% of commutes fall: $P(X < x) = 0.10$ and $P(X < x) = 0.80$.

b) **(2) Higher-than cutoffs.** If exam scores follow $X \sim N(72, 9^2)$, find the cutoff above which the highest 10% of scores fall and the cutoff above which the highest 80% of scores fall: $P(X > x) = 0.10$ and $P(X > x) = 0.80$.

c) **(3) Central band.** If battery life follows $X \sim N(20, 2.5^2)$, find the interval $[a, b]$ containing the middle 95% of battery lives: $P(a < X < b) = 0.95$.

d) **(3) Tail cutoffs.** If metal rod diameters follow $X \sim N(10.00, 0.08^2)$, find the two cutoff values a and b so that 10% of values are outside the interval: $P(X < a \text{ or } X > b) = 0.10$.

a) Lower-than cutoffs

Morning commute times follow

$$X \sim N(38, 7^2)$$

First cutoff. Find x such that $P(X < x) = 0.10$.

$$P(X < x) = 0.10 \qquad \longrightarrow \qquad \Phi(z) = 0.10$$

$$z = -1.282$$

$$x = \mu + \sigma z = 38 + 7(-1.282) = 29.026$$

$$x \approx 29.03$$

Second cutoff. Find x such that $P(X < x) = 0.80$.

$$P(X < x) = 0.80 \qquad \longrightarrow \qquad \Phi(z) = 0.80$$

$$z = 0.842$$

$$x = \mu + \sigma z = 38 + 7(0.842) = 43.894$$

$$x \approx 43.89$$

This inverse step depends on the same left-tail probability interpretation practiced in C5 part a).

b) Higher-than cutoffs

Exam scores follow

$$X \sim N(72, 9^2)$$

First cutoff. Find x such that $P(X > x) = 0.10$.

$$P(X > x) = 0.10 \qquad \longrightarrow \qquad P(X < x) = 0.90$$

$$\Phi(z) = 0.90 \qquad \longrightarrow \qquad z = 1.282$$

$$x = \mu + \sigma z = 72 + 9(1.282) = 83.538$$

$$x \approx 83.54$$

Second cutoff. Find x such that $P(X > x) = 0.80$.

$$P(X > x) = 0.80 \qquad \longrightarrow \qquad P(X < x) = 0.20$$

$$\Phi(z) = 0.20 \qquad \longrightarrow \qquad z = -0.842$$

$$x = \mu + \sigma z = 72 + 9(-0.842) = 64.422$$

$$x \approx 64.42$$

This inverse step depends on the same complement-rule interpretation practiced in C5 part b).

c) Central band

Battery life follows

$$X \sim N(20, 2.5^2)$$

Find the interval $[a, b]$ such that $P(a < X < b) = 0.95$.

$$1 - 0.95 = 0.05 \qquad \longrightarrow \qquad \frac{0.05}{2} = 0.025$$

$$P(X < a) = 0.025 \qquad P(X < b) = 0.975$$

$$z_a = -1.960 \qquad z_b = 1.960$$

$$a = \mu + \sigma z_a = 20 + 2.5(-1.960) = 15.100$$

$$b = \mu + \sigma z_b = 20 + 2.5(1.960) = 24.900$$

$$[a, b] = [15.10, 24.90]$$

This central interval uses the same interval and tail-area reasoning practiced in C5 part c).

d) Tail cutoffs

Metal rod diameters follow

$$X \sim N(10.00, 0.08^2)$$

Find a and b such that $P(X < a \text{ or } X > b) = 0.10$.

$$P(X < a \text{ or } X > b) = 0.10 \qquad \longrightarrow \qquad \frac{0.10}{2} = 0.05$$
$$P(X < a) = 0.05 \qquad P(X < b) = 0.95$$
$$z_a = -1.645 \qquad z_b = 1.645$$
$$a = \mu + \sigma z_a = 10.00 + 0.08(-1.645) = 9.8684$$
$$b = \mu + \sigma z_b = 10.00 + 0.08(1.645) = 10.1316$$
$$[a, b] \approx [9.87, 10.13]$$

This tail-cutoff problem uses the same interval and tail-area reasoning practiced in C5 part c).

8 C5 C6 C7 Minimal solutions

8.1 C5 Minimal solutions

a)

$$P(Z < 1.50) = 0.9332 \quad P(Z < -1.20) = 0.1151 \quad P(Z < 0.25) = 0.5987$$
$$P(Z < 1.15) = 0.8749 \quad P(Z < -1.05) = 0.1469 \quad P(Z < 1.35) = 0.9115$$

b)

$$P(Z > 1.50) = 0.0668 \quad P(Z > -0.80) = 0.7881 \quad P(Z > 1.35) = 0.0885$$

c)

$$P(0.25 < Z < 1.15) = 0.2762 \qquad P(Z < -1.05 \text{ or } Z > 1.35) = 0.2354$$
$$P(Z < -1.28) = 0.1003 \qquad P(Z < 0.84) = 0.7995$$
$$P(Z > 1.28) = 0.1003 \qquad P(Z > -0.84) = 0.7995$$
$$P(-1.96 < Z < 1.96) = 0.9500 \qquad P(Z < -1.64 \text{ or } Z > 1.64) = 0.1010$$

8.2 C6 Minimal solutions

a)

$$X \sim N(70, 8^2): \quad P(X < 82) = 0.9332 \quad X \sim N(70, 8^2): \quad P(X < 60.4) = 0.1151$$

b)

$$X \sim N(56, 5^2): \quad P(X > 63.5) = 0.0668 \quad X \sim N(56, 5^2): \quad P(X > 52) = 0.7881$$

c)

$$X \sim N(14, 4^2): \quad P(15 < X < 18.6) = 0.2762$$

d)

$$X \sim N(300, 20^2): \quad P(X < 279 \text{ or } X > 327) = 0.2354$$

8.3 C7 Minimal solutions

a)

$$X \sim N(38, 7^2): \quad P(X < x) = 0.10 \longrightarrow x \approx 29.03 \quad X \sim N(38, 7^2): \quad P(X < x) = 0.80 \longrightarrow x \approx 43.89$$

b)

$$X \sim N(72, 9^2): \quad P(X > x) = 0.10 \longrightarrow x \approx 83.54 \quad X \sim N(72, 9^2): \quad P(X > x) = 0.80 \longrightarrow x \approx 64.42$$

c)

$$X \sim N(20, 2.5^2): \quad P(a < X < b) = 0.95 \longrightarrow [a, b] = [15.10, 24.90]$$

d)

$$X \sim N(10.00, 0.08^2): \quad P(X < a \text{ or } X > b) = 0.10 \longrightarrow [a, b] \approx [9.87, 10.13]$$